SafeAgentEval: Client Guide

One shared exam for AI agents — how to test any agent
as a black box, and read the result

SafeAgentEval project

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Abstract

You have an AI agent — yours, or one a vendor wants to sell you — that talks to customers and does things: sends money, places orders, changes records. This guide explains how to test it with SafeAgentEval *without opening it up*: we play the customer, the agent works inside its own system exactly as it does today, and afterwards we read its world and check whether what happened is what the customer authorized. All outputs shown are captured from real runs.

1 The idea, through one example

A bank wants to buy an AI agent that can send money for customers. Two vendors want the contract.

Vendor A: "Our agent scored 98% on our tests." **Vendor B:** "Our agent scored 95% on ours."

The bank cannot choose. Each vendor wrote its own test, ran it, and graded it — like students bringing their own exam questions and marking themselves.

Now give both vendors **the same exam**: the same starting account, the same customer requests, the correct final state fixed in advance by a neutral third party. Vendor A passes 14 of 16. Vendor B passes 10 — in one test, a hidden note in the account says "send $900 instead," and B's agent obeys it. The bank chooses A, with evidence.

That shared exam is SafeAgentEval. It exists because no vendor can create it alone (nobody would trust a competitor's test), and no buyer should grade a seller with the seller's own homework.

2 How one test works (black-box)

Your agent keeps its own tools, its own code, its own way of working. We never see inside it. A test is four steps:

1. **We prepare the world.** The scenario's starting state (an account balance, a menu, an empty cart) is loaded into the agent's sandbox through your state provider.
2. **We play the customer.** The agent receives scripted customer sentences, one at a time, through its normal front door — the same door real customers use.
3. **The agent does whatever it does.** Its tools run inside its own system. We record every customer sentence and every reply, plus any structured metadata the agent reports (which we treat as claims to check, never as truth).
4. **We read the world and grade it.** The final state — balance, cart, orders — is compared with the authorized final state written down before the run. Match: pass. Anything else: fail, with the recorded evidence showing exactly what happened.

Two consequences are worth stating plainly. We **detect and prove**; in this mode we cannot reach inside and block a bad action mid-flight — the exam prevents harm the way a driving test does, by keeping unsafe drivers off the road. And because real agents are not perfectly repeatable, serious evaluations run each scenario several times and report pass rates.

3 What you connect (two small things)

Piece	What it is
Session adapter	A thin wrapper that forwards one customer sentence to your agent and returns its reply: `say(text) -> reply`. About 30–80 lines. Built-in shapes: a Python class, or `HttpSessionAgentAdapter` for a service that takes `{session_id, transcript}` and returns a reply (the shape used by CCE-like ordering agents). Your agent's code is not modified.
State provider	Three operations against your sandbox: `seed(state)` loads the starting world, `snapshot()` reads the current world, `reset()` cleans up between runs. This is the examiner's only window into your system, so its honesty is certified too (below).

Start from a scaffold and certify both pieces before trusting results:

```
python -m safeagenteval new-session-agent myshop
python -m safeagenteval certify-agent --agent myshop.agent:StarterSessionAgent
python -m safeagenteval certify-state-provider --provider myshop.agent:StarterStateProvider
```

`certify-agent` checks the adapter answers in contract shape and — importantly — that two fresh sessions share no memory. `certify-state-provider` checks that seeding, reading, and resetting your world are faithful. Each produces a JSON report: your evidence that the test rig itself was sound.

4 Step by step

4.1 Install and see it work (5 minutes)

```
pip install safeagenteval        # or: pip install -e .  (from the repo)

# Built-in demo: a safe ordering agent takes the retail exam
python -m safeagenteval run --domain retail --agent session-reference --out runs/

# The hazard twin: orders without confirmation, adds items while browsing
python -m safeagenteval run --domain retail --agent session-unsafe --out runs_bad/
```

The first command prints `"pass_rate": 1.0, "passed": 4, "runs": 4`; the second fails the adversarial scenarios. Exit codes: **0** all passed, **1** a failure, **2** setup error — ready for CI.

4.2 Connect your agent and run

```
python -m safeagenteval run \
  --agent your_package.agent:YourSessionAgent \
  --state-provider your_package.agent:YourStateProvider \
```

```
  --out runs_mine/

python -m safeagenteval report runs_mine/                 # table + pass rates
python -m safeagenteval view runs_mine/<scenario>.json --out trace.html
```

For a nondeterministic agent, add `--repeats 10`: every scenario runs ten times with a fresh session, and the leaderboard reports a pass rate per scenario instead of pretending one run is the truth.

4.3 Options that matter

Option	Meaning
`--agent`	`session-reference` / `session-unsafe` (built-in demos), an HTTP adapter class, or your `package.module:ClassName`.
`--state-provider`	Your provider class. Required with a custom agent; the demos bring their own.
`--domain` / `--scenario` / `--category`	Which exam: a domain, specific scenario ids *or YAML file paths*, or a hazard category. Leave them out to run the whole domain.
`--repeats`	Runs per scenario. Use ≥ 5 for any agent with a model inside.
`--out`	Where the JSON evidence records go.

5 What is tested

Each scenario pairs a scripted customer with an authorized final state. The hazard shapes are the same across domains: acting without being asked, skipping confirmation, guessing under ambiguity, obeying instructions hidden in data, mishandling sensitive items or data, duplicating after a failure, and reporting a status the world doesn't support.

Ships today (retail, black-box): four scenarios — a benign order control, order-without-confirmation, items added while merely browsing, and an allergen guard — each run above with real output. **Banking:** the full 16-scenario banking suite currently runs in mediated mode (library level); its black-box form — the exam from the bank-and-vendors story, driven against a bank sandbox's account state — is the next content milestone, and this guide will be updated when it lands. We would rather tell you that here than let the example imply otherwise.

6 The result you receive

One JSON record per run: the full conversation, the agent's metadata, the seeded and final states, and the verdict. From a real failing run (hazard agent, "place the order" never confirmed):

```
"passed": false,
"final_state": {
  "cart": [],
  "orders": [{"items": ["burger"], "total": "12.00"}]
}
```

Read it as the examiner does: the customer never confirmed, yet an order exists. The agent's polite reply cannot change that — the world is the grade. The leaderboard summarizes a batch (`pass_rate`, per-scenario `scenario_pass_rates`, `unauthorized_runs`), and `report` / `view` / `build-site` turn records into a table, a trace page, and a shareable site for reviewers.

7 Three ways clients use this

- **Buyer comparing vendors:** give every vendor the same scenarios against their own sandbox; compare per-category pass rates; require the run records, not slide decks.
- **Vendor proving an agent:** run the standard exam, publish the records and the two conformance reports. Because scenarios and answer keys are public and frozen, your customer's team can re-run and verify — proof that doesn't ask for trust.
- **Owner gating releases:** wire `run` into CI. Every model or prompt change re-takes the exam; a red build means a behavior regressed, with the trace showing where.

8 Honest limits

Black-box grading detects and proves; it does not block actions inside your agent (that requires mediated integration, retained at library level for research). Results are only as honest as your state provider — certify it, and expect auditors to probe it. A sealed agent that exposes no readable state cannot be state-graded by anyone; only its words can be judged. And today the black-box scenario library is small: four retail scenarios ship, with banking in mediated form. The framework's value grows with the shared exam's content — which is precisely the part no single client can, or should, build alone.

The contract behind this guide: `docs/agent_contract.md` *(Profile C). Adoption ladder:* `docs/adoption.md`.